

1. Topic Title

“Design and Implementation of an Ethical AI-Based Medical Chatbot for Non-Diagnostic Health Guidance and Preventive Care”

Topic Explanation (Simple Language)

This project focuses on building a **safe and ethical medical AI chatbot** that:

- Provides **health information in simple language**
- Offers **non-diagnostic symptom guidance**
- Gives **age-aware and gender-aware preventive suggestions**
- Uses **limited health history** to improve relevance
- Always encourages **doctor consultation when required**

The chatbot **supports users**, but **never replaces doctors**.

2. Research Gap

Despite the rapid growth of medical chatbots, several **critical gaps** still exist:

2.1 Ethical Personalization Gap

Existing medical chatbots fall into two extremes:

-  Fully generic (same response for all users)
-  Over-personalized (unsafe diagnosis, privacy risks)

Research Gap Identified:

There is a lack of **controlled, ethical personalization** that considers:

- Age
 - Gender
 - Basic health history
- without diagnosing diseases or violating privacy.**

2.2 Preventive Care Personalization Gap

Most systems:

- Give **general health**
- Ignore age- and gender-specific risks

Examples:

- Osteoporosis risk in elderly women
- Heart disease risk in middle-aged men

Research Gap Identified:

Insufficient focus on **age-aware and gender-aware preventive healthcare guidance**.

2.3 Personalization Limitation

- Current safe systems:
 - Give **same answers to all users**
 - Ignore age, gender, or medical
- Advanced systems:
 - Track sensitive personal data (privacy risk)

Gap Identified:

No balanced approach that allows **controlled personalization** (age, gender, history) **without violating privacy or ethics**.

2.4 Context Retention (History-Based Chat) Gap

Current systems:

- Either store **full medical records** (privacy risk)
- Or **forget everything** (poor user experience)

Research Gap Identified:

No effective use of **short-term, session-based health history** to improve response relevance **without long-term data storage**.

2.5 Accessibility & Language Gap

- Most systems use:
 - Complex medical terminology
 - English-heavy clinical language
- Not suitable for:
 - Rural users
 - Non-medical backgrounds

Gap Identified:

Limited availability of **simple-language, layman-friendly medical AI assistants**.

3. Implementation Criteria

3.1 Ethical & Safety Rules (Highest Priority)

- Non-diagnostic only
- Clear medical disclaimers
- Emergency symptom detection
- Doctor consultation prompts
- No treatment or dosage decisions

3.2 Functional Implementation Criteria

A. Health Information Module

- Explain diseases in **simple words**
- Example:
 -  “Hypertension is a chronic cardiovascular disorder”
 -  “High blood pressure means the force of blood against your arteries is too high”

B. Age-Aware Health Suggestions

The chatbot adjusts **preventive advice**, not diagnosis, based on age group.

Examples :

Age Group	Personalized Suggestions
Children	Nutrition, vaccination reminders
Young Adults	Mental health, fitness, sleep
Middle-aged	Blood pressure, diabetes prevention
Elderly	Fall prevention, bone health

⚠ No disease confirmation, only **risk awareness**.

C. Gender-Aware Preventive Tips

Gender is used **only for preventive guidance**, not diagnosis.

Examples :

Gender	Preventive Focus
Female	Anemia awareness, bone health
Male	Heart health, stress management
All	Mental health, hygiene, fitness

- ✓ Gender input is **optional**
- ✓ No sensitive conclusions drawn

C. History-Based Chat (Safe Context Memory)

The chatbot remembers **limited health-related context** within a session or opt-in history.

Allowed History Data

- ✓ Chronic conditions (user-stated)
- ✓ Allergies (optional)
- ✓ Lifestyle habits
- ✓ Past symptoms (short-term only)

Not Allowed

- ✗ Diagnosis
- ✗ Lab reports
- ✗ Medical records
- ✗ Long-term storage without consent

Example Interaction

User: *"I have diabetes"*

Chatbot:

"Since you mentioned diabetes earlier, here are general lifestyle tips that may help. Please consult your doctor for personalized treatment."

3.3 Functional Implementation Criteria

1. Symptom Guidance Engine

- Uses NLP to understand symptoms
- Provides **possible explanations**
- Adds age/gender risk awareness
- Triggers doctor visit advice when needed

2. Preventive Healthcare Module

- Age-based lifestyle tips
- Gender-based health awareness

3. Medication Information Module

- Purpose of medicine
- Side effects

3.4 Technical Architecture

Component	Role
UI Layer	User-friendly chatbot
NLP Engine	Symptom & intent detection
Personalization Engine	Age, gender, history logic
Safety Filter	Medical boundary enforcement
Knowledge Base	Verified medical data
Verified medical data	Session-based health context

3.5 Privacy & Data Protection

- ✓ User consent required
- ✓ No identity tracking
- ✓ Session-based memory
- ✓ Secure anonymized storage

4. One-Line Enhanced Project Summary

This project develops an ethical, personalized AI medical chatbot that provides age-aware, gender-aware, and history-based preventive health guidance while maintaining strict non-diagnostic medical boundaries.